

CLOSED-FACTOR MINIMA AND THE CLOSED-RICH CONSTANT OF THE FIBONACCI WORD

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ABSTRACT. Let f be the infinite Fibonacci word and let

$$M_n = \min\{\text{Cl}(w) : w \in \text{Fac}(f), |w| = n\},$$

where $\text{Cl}(w)$ is the number of distinct closed factors of w (including the empty word). We determine M_n for every n . In particular, we prove the first-difference formula conjectured by Maity and Puzynina and obtain the exact closed-rich constant

$$C_f = \frac{\phi^3}{(\phi^3 + 2)^2} = \frac{5\phi + 3}{\phi^2(\phi^3 + 2)^2}, \quad \phi = \frac{1 + \sqrt{5}}{2}.$$

For $k \geq 5$, at the lengths $F_k + 2F_{k-3} - 2$ the minimizing factor is unique and equals $f_{k-3}f_k f_{k-3}^-$. Singular kernels determine the repeated suffixes, and reversal symmetry reduces the minimization to a discrete concavity argument.

1. INTRODUCTION

A finite word is *closed* if its length is at most one, or if it has a nonempty proper border occurring exactly twice, as prefix and suffix. Occurrences may overlap. Write $\text{Cl}(w)$ for the number of distinct closed factors of w , including the empty word. For the Fibonacci word f , put

$$M_n = \min\{\text{Cl}(w) : w \in \text{Fac}(f), |w| = n\}, \quad M_0 = 1, \quad C_f = \inf_{n \geq 1} \frac{M_n}{n^2}.$$

Maity and Puzynina conjectured the first differences of M_n and the exact value of C_f [2, Conjectures 5.2–5.3]. Their Remark 5.4 also identifies a conjecturally unique minimizing factor at certain lengths. We prove these assertions. We use the singular-kernel results of Huang and Wen [1]; the closed-factor counts are derived below.

Let $f_{-1} = b$, $f_0 = a$ and $f_j = f_{j-1}f_{j-2}$ for $j \geq 1$, and let $f = \lim_j f_j$. Write $F_j = |f_j|$, so

$$F_{-1} = F_0 = 1, \quad F_{j+1} = F_j + F_{j-1}.$$

Throughout a Fibonacci block we use

$$(1) \quad \begin{aligned} k \geq 5, \quad A = F_{k-4}, \quad B = F_{k-3}, \quad C = A + B, \quad D = A + 2B, \quad K = 2A + 3B = F_k, \\ P = B - A = F_{k-5} > 0, \quad E = 2A - B = F_{k-6} > 0. \end{aligned}$$

In particular $A \geq 2$, $A < B < 2A$, $C = F_{k-2}$ and $D = F_{k-1}$. Empty integer ranges are always omitted. We write w^R for reversal, w^- for deletion of the last letter, and ^-w for deletion of the first letter. Prefixes and suffixes of length zero are empty.

Our main formula is the following.

Theorem 1.1. *Let $k \geq 5$ and $0 \leq a \leq D - 1$. Write*

$$(2) \quad m_k = 1 + B(2A + B + 2).$$

Then

$$(3) \quad M_{K+a} = \begin{cases} m_k + Aa - 2P, & 0 \leq a \leq A - 2, \\ m_k + Ca - A(B - 2), & A - 1 \leq a \leq B - 2, \\ m_k + Ba, & B - 1 \leq a \leq 2B - 2, \\ m_k + B(2B - 2) + D(a - 2B + 2), & 2B - 1 \leq a \leq D - 2, \\ m_k + C(D - 1) - AB + A, & a = D - 1. \end{cases}$$

Since $[F_k, F_{k+1} - 1] = [K, K + D - 1]$, these blocks cover all sufficiently large lengths without overlap. Taking differences gives the conjectured sequence.

Corollary 1.2. *For $k \geq 5$,*

$$(4) \quad R_{F_k+a} := M_{F_k+a} - M_{F_k+a-1} = \begin{cases} A, & 0 \leq a \leq A - 2, \\ C, & A - 1 \leq a \leq B - 2, \\ B, & B - 1 \leq a \leq 2B - 2, \\ D, & 2B - 1 \leq a \leq D - 2, \\ B, & a = D - 1. \end{cases}$$

Equivalently, if $v^{[r]}$ means r consecutive copies of the integer v , then $(R_n)_{n \geq 1}$ is the concatenation

$$(1, 1, 1, 1) \prod_{j=0}^{\infty} (F_j^{[F_j]}, F_{j+2}^{[F_{j-1}]}, F_j^{[F_j]}, F_{j+2}^{[F_{j-1}]}).$$

This is Conjecture 5.2 of [2].

Corollary 1.3. *For every $k \geq 5$, at*

$$N_k = F_k + 2F_{k-3} - 2$$

the minimum is attained by exactly one factor. It is

$$(5) \quad P_k = f_{k-3} f_k f_{k-3}^-, \quad M_{N_k} = 1 + F_{k-3} F_k.$$

Here w^- denotes deletion of the last two letters of w .

Theorem 1.4. *The closed-rich constant of the Fibonacci word is*

$$C_f = \frac{\phi^3}{(\phi^3 + 2)^2} = \frac{5\phi + 3}{\phi^2(\phi^3 + 2)^2}.$$

Thus Conjecture 5.3 of [2] holds.

2. SINGULAR KERNELS AND REPEATED SUFFIXES

2.1. The singular envelope. If α_j is the last letter of f_j , define

$$s_j = \alpha_{j-1} f_j^- \quad (j \geq 0), \quad s_{-1} = a.$$

Thus $s_{-1} = a$, $s_0 = b$, $s_1 = aa$, $s_2 = bab$. These words are palindromes, and $|s_j| = F_j$. The identities

$$(6) \quad s_j = s_{j-2} s_{j-3} s_{j-2}, \quad s_j^- = s_{j-2} s_{j-1}^- \quad (j \geq 2)$$

follow from the Fibonacci recursion; see also [1, Lemma 3.1].

The singular kernel of a factor is the longest singular word occurring in it; ties in length are resolved by taking the larger index. We use the following precise form of the singular-kernel description [1, Proposition 1.7, Theorem 2.1 and Corollary 3.4]. For $j \geq 1$, a factor with kernel s_j has a unique expression us_jv , with

$$0 \leq |u|, |v| < F_{j+1}.$$

The two extensions are the corresponding suffix and prefix of s_{j+1} . Every pair of lengths in this range is allowed, and the occurrences of us_jv correspond in order to the occurrences of s_j . Indeed, the common envelope is

$$(^-s_{j+1})s_j(s_{j+1}^-),$$

and its kernel is s_j , occurring once. For $j = 0$ the envelope is aba : the four words with designated kernel b are b, ab, ba, aba , and every b is flanked by a 's. For $j = -1$ the only word is a . These two observations extend the same statement to the short cases. In the one-letter tie, the larger index is chosen.

Positions start at zero. If $q_{j,p}$ is the start of the p th occurrence of s_j , then

$$(7) \quad q_{j,p} = pF_{j+1} + b_p F_j - 1,$$

where b_p counts the letters a in the first $p - 1$ letters of f . To see this, the first occurrence starts at $F_{j+1} - 1$, and the successive start-to-start distances are F_{j+2}, F_{j+1} in Fibonacci order [1, Proposition 1.7]. For $j = -1, 0$ the same conclusion follows from the substitution $a \mapsto ab, b \mapsto a$. Only the following values will be needed:

$$(8) \quad (b_1, \dots, b_{16}) = (0, 1, 1, 2, 3, 3, 4, 4, 5, 6, 6, 7, 8, 8, 9, 9).$$

They are the cumulative counts in the prefix $abaabababababab$.

Lemma 2.1 (Suffix intervals). *Fix an endpoint R and an occurrence $q = q_{j,p}$. If*

$$0 \leq R - q - F_j + 1 < F_{j+1},$$

then the suffixes ending at R with this designated kernel have exactly the lengths

$$(9) \quad R - q + 1 \leq \ell \leq R - q + F_{j+1}.$$

Let $d = q_{j,p} - q_{j,p-1}$ when $p > 1$, and let $d = \infty$ when $p = 1$. Such a suffix repeats within the length- N window ending at R if and only if

$$(10) \quad \ell + d \leq N.$$

Proof. The right extension has length $R - q - F_j + 1$. Letting the left extension range from 0 to $F_{j+1} - 1$ gives (9). Synchronization places the preceding copy exactly d positions to the left. It lies in the window precisely when (10) holds. If that copy starts before the window, every earlier copy does too. $\square$

2.2. The counting increment. Let $\text{LRS}(w)$ be the longest suffix occurring at least twice in w , with $\text{LRS}(w) = \varepsilon$ if no nonempty suffix repeats; set $\text{LRS}(\varepsilon) = \varepsilon$. Write

$$\varrho(w) = |\text{LRS}(w)| - |\text{LRS}(\text{LRS}(w))|.$$

We include a proof of the increment identity of [2, Lemma 3.1].

Lemma 2.2. *If appending the last letter does not enlarge the alphabet, then*

$$(11) \quad \text{Cl}(w) - \text{Cl}(w^-) = \varrho(w).$$

If the last letter is new, the increment is one.

Proof. Put $t = \text{LRS}(w)$ and $z = \text{LRS}(t)$. A new factor must be a suffix of length greater than $|t|$. For each nonempty suffix v of t , take its complete return formed by its last two occurrences in w . This is a closed suffix with longest border v . It has length greater than $|t|$ exactly when v does not repeat in t , that is, when $|z| < |v| \leq |t|$. Conversely, every new closed suffix arises this way from its longest border: that border repeats in w , hence is a suffix of t . Distinct longest borders give distinct closed suffixes. There are $|t| - |z|$ choices. If the last letter is new, the only new closed factor is that letter. $\square$

3. LOCAL SUFFIX PROFILES

3.1. Three families. Fix the parameters in (1) and $0 \leq a \leq D - 1$. The length- $(K + a)$ factors are precisely

$$(12) \quad W_{a,i}^{21} = \text{suf}_i(s_{k-1})s_k \text{pre}_{a-i}(s_{k-1}), \quad 0 \leq i \leq a,$$

$$(13) \quad W_{a,i}^{22} = \text{suf}_i(s_k)s_{k-1} \text{pre}_{C+a-i}(s_k), \quad 0 \leq i \leq C + a,$$

$$(14) \quad W_{a,i}^{23} = \text{suf}_{a+i+1}(s_{k-1})s_{k-2} \text{pre}_{D-i-1}(s_{k-1}), \quad 0 \leq i \leq D - a - 2.$$

Here is a direct consequence of the envelope description that also checks the boundary $a = 0$. A kernel of order at most $k - 3$ allows length at most $F_{k-3} + 2F_{k-2} - 2 = K - 2$, and a kernel of order at least $k + 1$ is too long. The only orders are therefore $k, k - 1, k - 2$. Distributing the remaining length between the two allowed extensions gives exactly (12)–(14). For the first family, s_{k+1} has prefix and suffix s_{k-1} by (6). Uniqueness of the kernel makes the parametrizations injective and their images disjoint. Their cardinalities sum to $K + a + 1$. The same classification, independently of the parameters A, B , is valid at $k = 4$ and will be used for the initial cases.

Proposition 3.1. Put $L_a = \max(0, a - B + 1)$ and $\varrho_{rs}(a, i) = \varrho(W_{a,i}^{rs})$. Then

$$(15) \quad \varrho_{21}(a, i) = \begin{cases} \min(D, C + 1 + i), & i \leq a - B, \\ \max(B, i + 1), & i > a - B, \end{cases}$$

$$(16) \quad \varrho_{22}(a, i) = \begin{cases} C, & i < L_a, \\ B, & L_a \leq i \leq a, \\ \min(C, B + i + 1), & a < i < B + a + 1, \\ \min(C, i + 1), & i \geq B + a + 1, \end{cases}$$

and

$$(17) \quad \varrho_{23}(a, i) = \begin{cases} \max(B, \min(C, a + i + 2)), & i < B, \\ \max(A, a + i - B + 2), & B \leq i < C, \\ B, & i \geq C. \end{cases}$$

Proof. Write $L = |\text{LRS}(W)|$ and $T = |\text{LRS}(\text{LRS}(W))|$. Appendix A gives the suffix intervals from Lemma 2.1. Each row specifies an entire interval of lengths and its preceding return distance; the rows cover all lengths above the value of T obtained below. Thus no longer suffix is omitted. We apply (10) first with $N = K + a$ and then with $N = L$.

Family 21. Put $x = a - i$. The interval $[C + x, K + x - 1]$ has distance D , so its largest repeated length is $L = C + a$. It lies in that interval because $0 \leq i \leq a < D$. Every larger length either fails the return inequality or lies in the first-occurrence interval. If $x < B$, the distance- D interval cannot repeat inside length L , since $L - D < C + x$. The distance- B interval gives

$$T = \min(C + x - 1, A + a).$$

If $x \geq B$, the two distance- D intervals join to $[x, K + x - 1]$. They contribute through $a - B$ when $a - B \geq x$; immediately below them, length $x - 1$ repeats, with distance A or C . Thus

$$T = \max(x - 1, a - B).$$

Subtracting these two values from $C + a$ proves (15).

Family 22. Put $x = a - i$, so $-C \leq x < D$. If $x \geq 0$, the distance- K interval begins at $C + x$. It contributes length a precisely when $i \geq C$; otherwise the next interval contributes its upper endpoint $C + x - 1$. If $x < 0$, the highest finite-distance interval has distance C . Consequently

$$(18) \quad L = \begin{cases} \max(a, C + x - 1), & x \geq 0, \\ \min(D + a, K + x - 1), & x < 0. \end{cases}$$

For the second pass there are four cases.

If $x \geq B$, then $i < C$ and $L = C + x - 1$. The distance- D interval starts at x , whereas $L - D = x - B - 1 < x$; the interval ending at $x - 1$ has distance at most C . Hence $T = x - 1$ and $L - T = C$.

If $0 \leq x < B$, the distance- B interval gives $T = L - B$, except possibly when $i < C$ and $x < P$. In that case $L - B = A + x - 1$ is one position below this interval; the interval of distance P ends at precisely that length and supplies the return. If $x \geq P$, the adjacent distance- B interval already covers it. Thus $L - T = B$ throughout.

If $-B \leq x < 0$, the distance- C intervals join to $[C + x, K + x - 1]$. When $i \geq A$, they contain $L - C$ and $L - T = C$. When $i < A$, one has $x > -A$ and $L = D + a$; the distance- C intervals cannot repeat, and the distance- B interval ends at $C + x - 1$. Thus $T = C + x - 1$ and $L - T = B + i + 1$. The two cases give $\min(C, B + i + 1)$.

Finally, if $-C \leq x < -B$, then $L - C \leq D + x - 1$, below the distance- C interval. The distance- A interval repeats through its upper endpoint $D + x - 1$, giving

$$L - T = \min(D + a, K + x - 1) - (D + x - 1) = \min(i + 1, C).$$

These are the four lines of (16).

Family 23. Here $a + i \leq D - 2$. The distance- D interval contributes $C + a$ when $a + i \geq B - 1$. Otherwise $i < B$ and the distance- C interval ends at $D - i - 2$. For $i \geq C$ the highest finite-distance interval repeats through $K - i - 2$. Therefore

$$L = \begin{cases} \max(C + a, D - i - 2), & i < B, \\ C + a, & B \leq i < C, \\ K - i - 2, & i \geq C. \end{cases}$$

If $i < B$, the distance- D interval cannot repeat inside length L . The distance- C interval contributes a exactly when $a + i \geq C - 1$; otherwise the next interval, of distance B or P , ends at $C - i - 2$ and repeats. Hence

$$T = \max(a, C - i - 2).$$

Putting $z = a + i + 2$, subtraction gives $L - T = \max(C + z, D) - \max(z, C) = \max(B, \min(C, z))$. If $B \leq i < C$, the distance- A interval gives

$$T = \min(B + a, D - i - 2), \quad L - T = \max(A, a + i - B + 2).$$

If $i \geq C$, the length $2C - i - 2$ lies in the distance- B interval when $i < 2A + B$, and at the upper endpoint of the distance- P interval otherwise. It repeats in both cases; the next length cannot repeat. Thus $T = 2C - i - 2$ and $L - T = B$, completing the proof. $\square$

4. REVERSAL, DISCRETE CONCAVITY, AND EXACT MINIMA

4.1. A minimization lemma. The Fibonacci word contains neither bb nor aaa . Consequently every factor of length at least three has both letters, and all one-letter deletions in this section preserve the alphabet. Fix one of the three families, suppress a and the family label, and write

$$G(i) = \text{Cl}(W_{a,i}), \quad r(i) = \varrho(W_{a,i}), \quad 0 \leq i \leq h.$$

The three values of h are a , $C + a$, and $D - a - 2$.

Since every singular word is a palindrome, the parametrizations (12)–(14) give

$$(19) \quad W_{a,i}^R = W_{a,h-i}, \quad G(i) = G(h-i).$$

Moreover $W_{a,i}^- = W_{a,i+1}$. Applying Lemma 2.2 once on each side of this common length- $(K + a - 1)$ word, and using (19), yields

$$(20) \quad \boxed{G(i+1) - G(i) = r(h-i-1) - r(i)}.$$

Lemma 4.1 (Discrete concavity). *Let J be the set of downward-jump indices of r ,*

$$J = \{j : 1 \leq j \leq h, r(j) < r(j-1)\}.$$

Then the minimum of G is attained among

$$(21) \quad \{0, h\} \cup J \cup \{h-j : j \in J\}.$$

Proof. Away from an index in J and its reflection, both increments appearing below are nonnegative:

$$\begin{aligned} & [G(i+1) - G(i)] - [G(i) - G(i-1)] \\ &= -[r(h-i) - r(h-i-1)] - [r(i) - r(i-1)] \leq 0. \end{aligned}$$

Thus G is discretely concave on each interval cut out by the points in (21). A concave sequence takes its minimum on an interval at an endpoint. $\square$

4.2. Baseline values. Set

$$m = \text{Cl}(s_k).$$

From (15), extending $W_{a,0}^{21} = s_k \text{pre}_a(s_{k-1})$ one letter at a time gives

$$(22) \quad H_{21}(a) := \text{Cl}(W_{a,0}^{21}) = m + B \min(a, B-1) + (C+1)(a-B+1)_+.$$

To anchor the 22 family, let $v = f_k^-$ and let α, β be the last letters of f_k, f_{k-1} , respectively. Then

$$W_{0,0}^{22} = s_{k-1}s_{k-2} = \alpha v, \quad s_k = \beta v.$$

By reversal, (16) at $i = C$ gives the increment C for prepending α , whereas (15) gives increment B for prepending β . Hence $\text{Cl}(W_{0,0}^{22}) = m + C - B = m + A$. Right extension now gives

$$(23) \quad H_{22}(a) = m + A + B \min(a, B-1) + C(a-B+1)_+.$$

Finally (6) gives $W_{a,0}^{23} = (W_{a+1,a+1}^{21})^-$, so

$$(24) \quad H_{23}(a) = H_{21}(a+1) - \max(B, a+2), \quad a \leq D-2.$$

These relations also determine m without a separate counting formula. Since $s_{k+1}^- = s_{k-1}s_k^- = W_{D-1,0}^{22}$ and $\varrho(s_{k+1}) = C$, we obtain

$$m_{k+1} = m_k + B^2 + C^2 + 2A.$$

Appendix B gives $\text{Cl}(s_5^-) = 25$, and $\varrho(s_5) = F_2 = 3$, so $m_5 = 28$. The expression

$$(25) \quad m_k = 1 + B(2A + B + 2)$$

has this initial value and the displayed recurrence, since the next pair (A, B) is (B, C) .

4.3. Minima in the three families. We apply Lemma 4.1. All candidate values below use the same telescoping identity, with $S(v) = \sum_{j=0}^{v-1} r(j)$:

$$(26) \quad G(i) - G(0) = S(h) - S(h-i) - S(i).$$

For clarity, each linear piece of S can be evaluated by

$$\sum_{j=u}^{v-1} (c+j) = (v-u)c + \frac{(v-u)(u+v-1)}{2}.$$

Splitting at the thresholds in (15)–(17) gives the candidate differences recorded below. Family 21. The only possible downward jump of (15) is at

$$u = a - B + 1.$$

If $a \leq B - 1$, then $r(i) = B$ throughout and G_{21} is constant. If $B \leq a \leq 2B - 2$, direct summation of (20) gives

$$(27) \quad G_{21}(u) - H_{21}(a) = -(A+1)u,$$

and G_{21} is constant from u to its reflection $B - 1$. If $a \geq 2B - 1$, the reflected candidate gives

$$(28) \quad G_{21}(B-1) - H_{21}(a) = (B-1)(a-D+1).$$

Consequently

$$(29) \quad \min_i G_{21}(i) = \begin{cases} m + Ba, & 0 \leq a \leq 2B - 2, \\ m + B(2B - 2) + D(a - 2B + 2), & 2B - 1 \leq a \leq D - 1. \end{cases}$$

The drop in (27) is strictly negative. Concavity therefore makes every point outside the central plateau strictly larger than its value. For $a = B - 1$ the whole family is constant. Thus, for $B - 1 \leq a \leq 2B - 2$, the complete minimizing interval is

$$(30) \quad a - B + 1 \leq i \leq B - 1.$$

Family 22. The only possible downward jumps are

$$L_a = \max(0, a - B + 1), \quad B + a + 1.$$

The second is a genuine downward jump only when $a \leq A - 3$; its reflection is $A - 1$, and

$$(31) \quad G_{22}(A-1) - H_{22}(a) = A(a+1) > 0.$$

For the first jump, which occurs when $a \geq B$,

$$(32) \quad G_{22}(L_a) - H_{22}(a) = \begin{cases} 0, & B \leq a \leq D - 2, \\ -A, & a = D - 1. \end{cases}$$

Thus

$$(33) \quad \min_i G_{22}(i) = \begin{cases} m + A + Ba, & 0 \leq a \leq B - 1, \\ m + Ca - AB + 2A, & B \leq a \leq D - 2, \\ m + C(D - 1) - AB + A, & a = D - 1. \end{cases}$$

Family 23. The only possible downward jumps of (17) are at B and C . From (24), for $a \leq B - 2$ one has $H_{23}(a) = m + Ba$. Whenever the indicated candidates lie in the parameter range, summing (20) gives

$$(34) \quad G_{23}(B) - H_{23}(a) = \begin{cases} -(a+2)(B-A), & 0 \leq a \leq A-3, \\ A(a-B+2), & A-2 \leq a \leq B-2, \end{cases}$$

and

$$(35) \quad G_{23}(C) = H_{23}(a) \quad (0 \leq a \leq B-2).$$

For $a \geq B-1$, (24) becomes

$$H_{23}(a) = m + Ca - AB + 2A,$$

and, if B remains in range, $G_{23}(B) = H_{23}(a)$. Lemma 4.1 therefore gives

$$(36) \quad \min_i G_{23}(i) = \begin{cases} m + Aa - 2P, & 0 \leq a \leq A-2, \\ m + Ca - AB + 2A, & A-1 \leq a \leq D-2. \end{cases}$$

4.4. Proof of the main counting theorem. Comparing (29), (33), and (36) is now elementary. We record the differences that determine the winner:

$$\begin{aligned} \min G_{22} - \min G_{21} &= A > 0, & 0 \leq a \leq A-2, \\ \min G_{21} - \min G_{23} &= P(a+2) > 0, & 0 \leq a \leq A-2, \\ \min G_{21} - \min G_{23} &= A(B-a-2) \geq 0, & A-1 \leq a \leq B-2, \\ \min G_{22} - \min G_{23} &= A(B-a-1) > 0, & A-1 \leq a \leq B-2, \\ \min G_{22} - \min G_{21} &= A(a-B+2) > 0, & B-1 \leq a \leq 2B-2, \\ \min G_{22} - \min G_{21} &= B(D-a-2) \geq 0, & 2B-1 \leq a \leq D-2. \end{aligned}$$

On $B-1 \leq a \leq D-2$, equations (33) and (36) give $\min G_{22} = \min G_{23}$ whenever the 23 family is present. Thus the displayed comparisons cover all three families on every interval. At $a = D-1$ the 23 family is empty and

$$\min G_{21} - \min G_{22} = A + B > 0.$$

This proves Theorem 1.1.

For $a \geq 1$, Corollary 1.2 follows by subtracting adjacent cases in (3). For $k \geq 6$, the preceding block is available and

$$M_{F_k} = 1 + 2AB + B^2 + 2A, \quad M_{F_{k-1}} = 1 + 2AB + B^2 + A,$$

where the second value is the last case of the preceding Fibonacci block. Hence $R_{F_k} = A$ for $k \geq 6$. At $k = 5$, $M_{13} = 28 - 2 = 26$ and Appendix B gives $M_{12} = 24$, so $R_{13} = 2 = A$ as well. The initial differences and the block formula concatenate into the sequence in Corollary 1.2.

At $a = 2B - 2$, the interval (30) shrinks to the single index $i = B - 1$, while both other families are larger by $AB > 0$. To identify this factor, let $g = f_{k-3}$ and $h = f_{k-4}$, and let α be the last letter of f_k and β the last letter of f_{k-1} . Then $f_{k-1} = ghg$, the last letter of g is β , and

$$s_{k-1} = \alpha ghg^-, \quad s_k = \beta f_k^-.$$

Hence

$$\text{suf}_{B-1}(s_{k-1}) = g^-, \quad \text{pre}_{B-1}(s_{k-1}) = \alpha g^{--},$$

and therefore, since $g = g^- \beta$ and $f_k = f_k^- \alpha$,

$$W_{2B-2, B-1}^{21} = g^- \beta f_k^- \alpha g^{--} = f_{k-3} f_k f_{k-3}^-.$$

Finally (25) and (29) give

$$M_{N_k} = m + B(2B - 2) = 1 + BK.$$

This proves Corollary 1.3.

The lengths below F_5 are given explicitly in Appendix B. Together with Theorem 1.1, this completes Conjecture 5.2 for every length.

5. THE CLOSED-RICH CONSTANT

We prove Theorem 1.4. Linearly interpolate M_n between successive integers and call the resulting function $M(x)$. On a segment where $M(x) = ux + v$, with $u > 0$,

$$\frac{d}{dx} \frac{M(x)}{x^2} = \frac{-ux - 2v}{x^3}.$$

The numerator is decreasing, so an interior critical point can only be a maximum. At a downward change of slope, the derivative also jumps downward. Thus, between consecutive upward changes of slope, the minimum of $M(x)/x^2$ occurs at an endpoint.

For $x \geq 13$, Corollary 1.2 shows that the upward changes occur exactly at

$$T_k = K + A - 2 = 3C - 2, \quad N_k = K + 2B - 2 \quad (k \geq 5).$$

There is no extra change at a Fibonacci boundary: the final slope B of one block equals the initial slope of the next. Since $T_5 = 13$, these intervals cover $[13, \infty)$. Theorem 1.1 gives

$$M_{T_k} = 1 + C^2, \quad M_{N_k} = 1 + BK.$$

The first ratios satisfy

$$(37) \quad \frac{M_{T_k}}{T_k^2} = \frac{1 + C^2}{(3C - 2)^2} > \frac{1}{9}.$$

Set $t = \phi^3 > 4$ and

$$q_k = \frac{1 + BK}{(K + 2B - 2)^2}, \quad \lambda = \frac{t}{(t + 2)^2}.$$

Because $(t + 2)^2 - 9t = (t - 1)(t - 4) > 0$, one has $\lambda < 1/9$. Binet's formula gives the exact identity

$$K = tB + e_k, \quad e_k = 2(-1)^{k-1}\phi^{-(k-1)}, \quad |e_k| < 1.$$

Clearing the denominators in $q_k - \lambda$ gives

$$\begin{aligned} & (1 + BK)(t + 2)^2 - t(K + 2B - 2)^2 \\ &= (t + 2)B(4t + (2 - t)e_k) + (t + 2)^2 - t(e_k - 2)^2. \end{aligned}$$

Both terms are positive: the bracket is greater than $3t + 2$, and

$$(t + 2)^2 - t(e_k - 2)^2 > (t + 2)^2 - 9t = (t - 1)(t - 4) > 0.$$

Thus $q_k > \lambda$ for every $k \geq 5$, while $q_k \rightarrow \lambda$ because $B \rightarrow \infty$ and $e_k \rightarrow 0$. Together with (37) and the interpolation argument, this proves $M_n/n^2 > \lambda$ for $n \geq 13$. The initial table gives $M_n/n^2 \geq 1/6$ for $1 \leq n \leq 12$. Taking the infimum yields $C_f = \lambda$. Finally $\phi^5 = 5\phi + 3$, giving the second form of the constant in Theorem 1.4.

APPENDIX A. SUFFIX INTERVALS

The following tables give the intervals used in Proposition 3.1. We realize each word using the first occurrence of its central kernel: s_k , s_{k-1} or s_{k-2} . Their starting positions are $K + D - 1$, $K - 1$ and $D - 1$, respectively. Adding the core and right extension lengths gives the endpoints R displayed below. A row $(j - k, p)$ refers to the occurrence $q_{j,p}$ in (7); d is its preceding start-to-start distance. The parameter conditions are intersected with the ranges in (12)–(14). The interval $[u, v]$ is inclusive. Only its intersection with the window lengths under consideration is relevant.

To derive any row, substitute $q = q_{j,p}$ in

$$[R - q + 1, R - q + F_{j+1}], \quad 0 \leq R - q - F_j + 1 < F_{j+1}, \quad d = q_{j,p} - q_{j,p-1}.$$

For example, in family 22, the row $(-6, 16)$ has $F_{k-6} = E$, $F_{k-5} = P$, and $q = 16P + 9E - 1 = 2A + 7B - 1$. Since $R = 2K + x - 2$, it gives $[E + x, A + x - 1]$, condition $0 \leq x < P$, and $d = P$. Thus the table entries follow by subtraction, including the low orders -1 and 0 covered by the letter envelopes in Section 2. The intervals in each applicable row join at successive integer lengths down to, and including, the second repeated suffix used in the proof.

Family 21. Here $x = a - i$, $0 \leq x < D$, and $R = 2K + D + x - 2$.

$j - k$	p	condition	length interval	d
0	1	all	$[K + x, K + x + F_{k+1} - 1]$	∞
-2	3	all	$[C + x, K + x - 1]$	D
-3	5	$B \leq x < D$	$[x, C + x - 1]$	D
-4	8	$0 \leq x < B$	$[A + x, C + x - 1]$	B
-5	14	$B \leq x < C$	$[x - A, x - 1]$	A
-4	9	$C \leq x < D$	$[x - B, x - 1]$	C

Family 22. Here $x = a - i$, $-C \leq x < D$, and $R = 2K + x - 2$.

$j - k$	p	condition	length interval	d
-1	1	all	$[K + x, 2K + x - 1]$	∞
-2	2	$0 \leq x < D$	$[C + x, K + x - 1]$	K
-3	4	$B \leq x < D$	$[x, C + x - 1]$	D
-3	3	$-C \leq x < 0$	$[D + x, K + x - 1]$	C
-4	7	$C \leq x < D$	$[x - B, x - 1]$	C
-4	6	$0 \leq x < B$	$[A + x, C + x - 1]$	B
-4	5	$-B \leq x < 0$	$[C + x, D + x - 1]$	C
-5	11	$B \leq x < C$	$[x - A, x - 1]$	A
-5	10	$P \leq x < B$	$[x, A + x - 1]$	B
-5	9	$-A \leq x < 0$	$[B + x, C + x - 1]$	B
-5	8	$-C \leq x < -B$	$[2B + x, D + x - 1]$	A
-6	16	$0 \leq x < P$	$[E + x, A + x - 1]$	P

Family 23. Here $0 \leq i \leq D - a - 2$ and $R = K + D - i - 3$.

$j - k$	p	condition	length interval	d
-2	1	all	$[K - i - 1, K + D - i - 2]$	∞
-3	2	$0 \leq i < C$	$[D - i - 1, K - i - 2]$	D
-4	4	$0 \leq i < B$	$[C - i - 1, D - i - 2]$	C
-4	3	$C \leq i < D$	$[2C - i - 1, K - i - 2]$	B
-5	7	$0 \leq i < A$	$[B - i - 1, C - i - 2]$	B
-5	6	$B \leq i < C$	$[2B - i - 1, D - i - 2]$	A
-5	5	$C \leq i < 2A + B$	$[D - i - 1, 2C - i - 2]$	B
-6	11	$A \leq i < B$	$[2A - i - 1, C - i - 2]$	P
-6	8	$2A + B \leq i < D$	$[3A + B - i - 1, 2C - i - 2]$	P

APPENDIX B. INITIAL COUNTS

At $k = 4$, $a = 4$, the classification in Section 3 gives five factors of family 21, eight of family 22, and none of family 23. Using $s_3 = aabaa$ and $s_4 = babaabab$ gives the thirteen words in the table. Every shorter factor extends to a length-12 factor, so their prefixes also exhaust the shorter lengths. Entries give $\text{Cl}(\text{pre}_n(w))$.

w	$n = 5$	6	7	8	9	10	11	12
aabaababaaba	6	9	12	14	16	19	22	25
aababaabaaba	7	9	10	13	16	19	22	25
aababaababaa	7	9	10	13	16	19	22	27
abaabaababaa	7	9	12	15	18	20	22	25
abaababaabaa	7	9	11	13	16	19	22	25
abaababaabab	7	9	11	13	16	19	22	27
ababaabaabab	8	9	11	13	16	19	22	24
ababaababaab	8	9	11	13	15	17	22	27
baabaababaab	7	9	12	15	17	19	22	25
baababaabaab	7	9	11	13	16	19	22	25
baababaababa	7	9	11	13	16	19	23	27
babaabaababa	7	9	11	14	17	20	22	24
babaababaaba	7	9	11	13	15	19	23	27

The entries follow by successive application of Lemma 2.2, starting from $\text{Cl}(\varepsilon) = 1$. For example, in the first row the lengths of the two successive repeated suffixes, for $n = 5, \dots, 12$, are respectively

$$(2, 3, 4, 2, 3, 4, 5, 6), \quad (1, 0, 1, 0, 1, 1, 2, 3).$$

The increments are therefore $(1, 3, 3, 2, 2, 3, 3, 3)$, giving that row from $\text{Cl}(aaba) = 5$. Every one-letter extension increases Cl by at least one, while $a, aa, aab, aaba$ have counts 2, 3, 4, 5. Thus

$$(M_0, \dots, M_{12}) = (1, 2, 3, 4, 5, 6, 9, 10, 13, 15, 17, 22, 24).$$

Also $s_5^- = aabaababaaba$, so the first row gives the base value used in Section 4.

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